

	so internal energy increases	A1	[1]
4	(a) mass per unit volume (<i>ratio idea must be clear, not units</i>)	B1	[1]
	(b) (i) pressure is same at the surface of <u>mercury</u> because at same horizontal level	B1	[1]
	(ii) $h\rho g$ is same for both	B1	
	$53 \times 10^{-2} \times 1.0 \times 10^3 \times g = 71 \times 10^{-2} \times \rho \times g$	C1	
	$\rho = 7.5 \times 10^2 \text{ kg m}^{-3}$	A1	[3]

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5 (a) no hysteresis loop/no permanent deformation

M1